

Start coding or [generate](#) with AI.

✓ **Project Name -**

Exploratory Data Analysis (EDA) on FedEx Logistics Supply Chain Delivery Dataset

✓ **Project Type -** EDA/Regression/Classification/Unsupervised

Contribution - Individual/Team

Team Member 1 -

Team Member 2 -

Team Member 3 -

Team Member 4 -

Write the summary here within 500-600 words.

> **GitHub Link -**

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✓ **Problem Statement**

FedEx Logistics manages thousands of shipments across multiple countries, vendors, and transportation modes. As shipment volume increases, monitoring delivery performance, transportation costs, and operational efficiency becomes increasingly challenging. The company needs a data-driven analysis to identify patterns affecting delivery performance, understand the factors influencing freight costs, compare shipment methods, evaluate vendor and manufacturing site performance, and detect opportunities to optimize logistics operations. The objective is to analyze historical shipment data and generate meaningful insights that support faster deliveries, cost optimization, and improved supply chain efficiency.

Write Problem Statement Here.

✓ **Define Your Business Objective?**

The primary objectives of this project are: Analyze shipment distribution across different countries. Evaluate the performance of various shipment modes. Study vendor and manufacturing site contributions. Analyze freight cost distribution and identify cost drivers. Examine the relationship

between shipment weight and freight cost. Evaluate delivery timelines and identify potential delays. Identify trends in product groups and shipment volumes. Detect anomalies, missing values, and inconsistencies in the dataset. Generate actionable business insights to improve supply chain efficiency and reduce logistics costs.

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✓ **General Guidelines : -**

1. Well-structured, formatted, and commented code is required.
2. Exception Handling, Production Grade Code & Deployment Ready Code will be a plus. Those students will be awarded some additional credits.

The additional credits will have advantages over other students during Star Student selection.

[Note: - Deployment Ready Code is defined as, the whole .ipynb notebook should be executed without a single error logged.]

3. Each and every logic should have proper comments.
4. You may add as many number of charts you want. Make Sure for each and every chart the following format should be answered.

Chart visualization code

- Why did you pick the specific chart?
- What is/are the insight(s) found from the chart?
- Will the gained insights help creating a positive business impact? Are there any insights that lead to negative growth? Justify with specific reason.

5. You have to create at least 20 logical & meaningful charts having important insights.

[Hints : - Do the Visualization in a structured way while following "UBM" Rule.

U - Univariate Analysis,

B - Bivariate Analysis (Numerical - Categorical, Numerical - Numerical, Categorical - Categorical)

M - Multivariate Analysis]

✓ ***Let's Begin !***

✓ ***1. Know Your Data***

Import Libraries

```
# Importing necessary libraries

import pandas as pd
import numpy as np

# Visualization libraries
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

# Ignore warnings
import warnings
warnings.filterwarnings('ignore')

# Set visualization style
plt.style.use('ggplot')
sns.set_theme(style='whitegrid')

print("Libraries imported successfully!")
```

Dataset Loading

```
# Load the dataset

df = pd.read_csv("SCMS_Delivery_History_Dataset.csv")

# Display first 5 rows
df.head()
```

```
import os
os.listdir()
```

Dataset First View

```
# Number of rows and columns

print("Shape of Dataset:", df.shape)
print("Number of Rows:", df.shape[0])
print("Number of Columns:", df.shape[1])
```

```
df.head()
```

```
df.tail()
```

Dataset Rows & Columns count

```
# Number of rows and columns

print("Shape of Dataset:", df.shape)
print("Number of Rows:", df.shape[0])
print("Number of Columns:", df.shape[1])
```

✓ Dataset Information

```
df.info()
```

✓ Duplicate Values

```
# Dataset Duplicate Value Count

duplicate_count = df.duplicated().sum()

print("Total Duplicate Rows:", duplicate_count)
```

✓ Missing Values/Null Values

```
# Missing Values Count

missing_values = df.isnull().sum()

missing_values
```

```
missing_percentage = (df.isnull().sum()/len(df))*100

missing_percentage.sort_values(ascending=False)
```

```
plt.figure(figsize=(14,6))
sns.heatmap(df.isnull(), cbar=False, cmap='viridis')
plt.title("Missing Values Heatmap")
plt.show()
```

✓ What did you know about your dataset?

Dataset Insights:

Observations

1. The dataset contains 10,324 rows and 33 columns.
2. No duplicate records were found in the dataset.
3. Most columns are complete; however, a few columns contain missing values.
4. Columns such as Shipment Mode, Dosage, and Line Item Insurance (USD) have missing values and will be handled during the data preprocessing stage.
5. The dataset contains numerical, categorical, and date-related variables.

6. The dataset is suitable for logistics and supply chain analysis, including shipment performance, freight cost, and delivery trends.

✓ 2. Understanding Your Variables

```
# Dataset Columns
```

```
# Dataset Describe
```

✓ Variables Description

The dataset consists of 33 variables related to FedEx logistics and supply chain operations. These variables include shipment identifiers, project details, destination country, shipment mode, vendor information, manufacturing site, product details, delivery dates, shipment weight, freight cost, and insurance cost. Together, these variables help analyze shipment performance, delivery efficiency, transportation costs, and supply chain operations.

✓ Check Unique Values for each variable.

```
df.nunique()
```

```
# Display unique values for each column

for column in df.columns:
    print(f"\n{'='*60}")
    print(f"Column Name: {column}")
    print(f"Number of Unique Values: {df[column].nunique()}")
    print(df[column].unique())
```

```
df.dtypes
```

Observation

- The dataset contains both categorical and numerical variables.
- Some columns have a large number of unique values, such as ID, Project Code, Vendor, and Item Description.
- Numerical columns include shipment quantity, shipment value, weight, freight cost, insurance cost, pack price, and unit price.
- Categorical columns include country, shipment mode, vendor, product group, manufacturing site, and brand.
- Date-related columns will be converted into datetime format during the Data Wrangling stage.

3. *Data Wrangling*

Data Wrangling Code

```
# Fill missing values in categorical columns with Mode

df['Shipment Mode'] = df['Shipment Mode'].fillna(df['Shipment Mode'].mode()[0])
df['Dosage'] = df['Dosage'].fillna(df['Dosage'].mode()[0])

# Fill missing values in numerical column with Median
df['Line Item Insurance (USD)'] = df['Line Item Insurance (USD)'].fillna(df['Line Item Insurance (USD)'].median())
```

```
df.isnull().sum()
```

What all manipulations have you done and insights you found?

Checked the dataset structure and data types.

Verified duplicate records and found no duplicate rows.

Identified missing values in Shipment Mode, Dosage, and Line Item Insurance (USD).

Replaced missing categorical values using the mode.

Replaced missing numerical values using the median.

Verified that the dataset contains no missing values after preprocessing.

Prepared the cleaned dataset for visualization and analysis.

4. *Data Vizualization, Storytelling & Experimenting with charts : Understand the relationships between variables*

Chart - 1

```
import matplotlib.pyplot as plt

top_countries = df['Country'].value_counts().head(10)

plt.figure(figsize=(10,6))
top_countries.plot(kind='bar')
plt.title('Top 10 Countries by Shipment Count')
plt.xlabel('Country')
plt.ylabel('Number of Shipments')
plt.xticks(rotation=45)
plt.show()
```

✓ 1. Why did you pick the specific chart?

A bar chart is ideal for comparing shipment counts across different countries.

✓ 2. What is/are the insight(s) found from the chart?

It identifies the countries receiving the highest number of shipments.

✓ 3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

The organization can prioritize logistics, inventory planning, and resource allocation for high demand countries.

✓ Chart - 2

```
plt.figure(figsize=(8,5))
df['Shipment Mode'].value_counts().plot(kind='bar')
plt.title('Shipment Mode Distribution')
plt.xlabel('Shipment Mode')
plt.ylabel('Count')
plt.show()
```

✓ 1. Why did you pick the specific chart?

To compare the usage frequency of different shipment modes.

✓ 2. What is/are the insight(s) found from the chart?

Shows which transportation method is used most frequently.

✓ 3. Will the gained insights help creating a positive business impact?

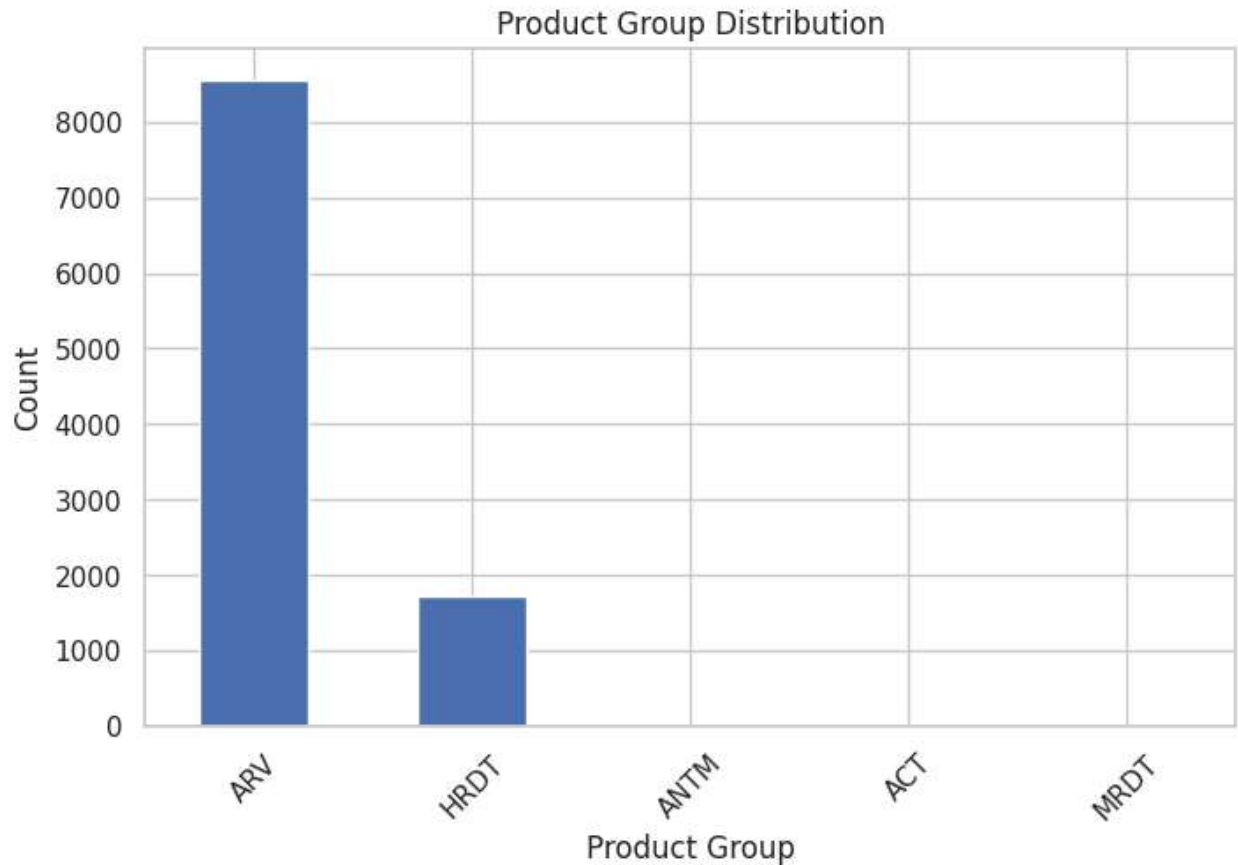
Are there any insights that lead to negative growth? Justify with specific reason.

Helps optimize transportation costs and improve logistics planning.

✓ Chart - 3

```
plt.figure(figsize=(8,5))
df['Product Group'].value_counts().plot(kind='bar')
plt.title('Product Group Distribution')
plt.xlabel('Product Group')
```

```
plt.ylabel('Count')
plt.xticks(rotation=45)
plt.show()
```



- ✓ 1. Why did you pick the specific chart?

To compare shipment volumes across product groups.

- ✓ 2. What is/are the insight(s) found from the chart?

Identifies the most frequently supplied product groups.

- ✓ 3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

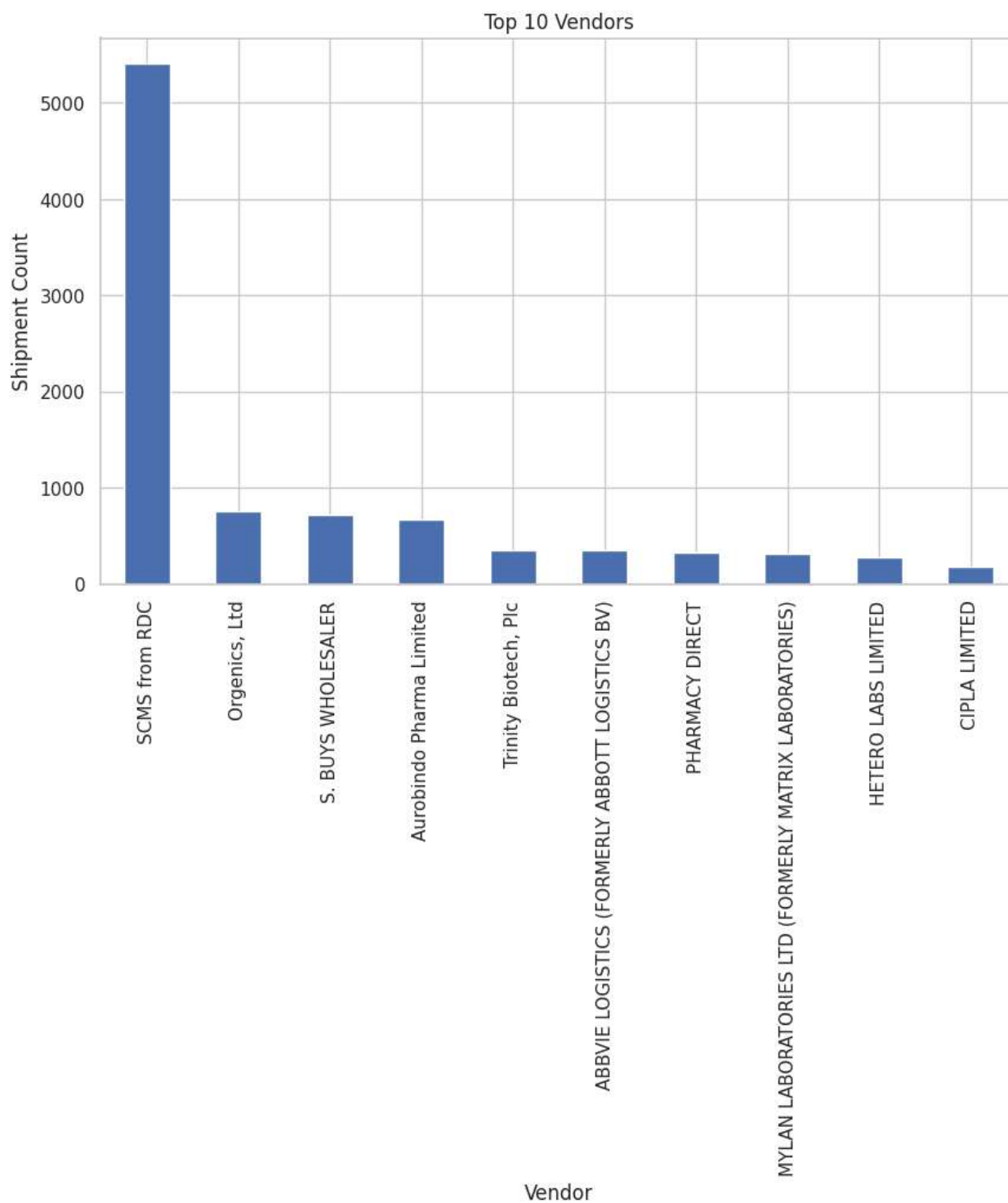
Supports inventory planning and procurement decisions.

- ✓ Chart - 4

```
top_vendor = df['Vendor'].value_counts().head(10)

plt.figure(figsize=(10,6))
```

```
top_vendor.plot(kind='bar')
plt.title('Top 10 Vendors')
plt.xlabel('Vendor')
plt.ylabel('Shipment Count')
plt.xticks(rotation=90)
plt.show()
```



✓ 1. Why did you pick the specific chart?

To identify the vendors supplying the highest number of shipments.

- ✓ 2. What is/are the insight(s) found from the chart?

Highlights the major suppliers in supply chain.

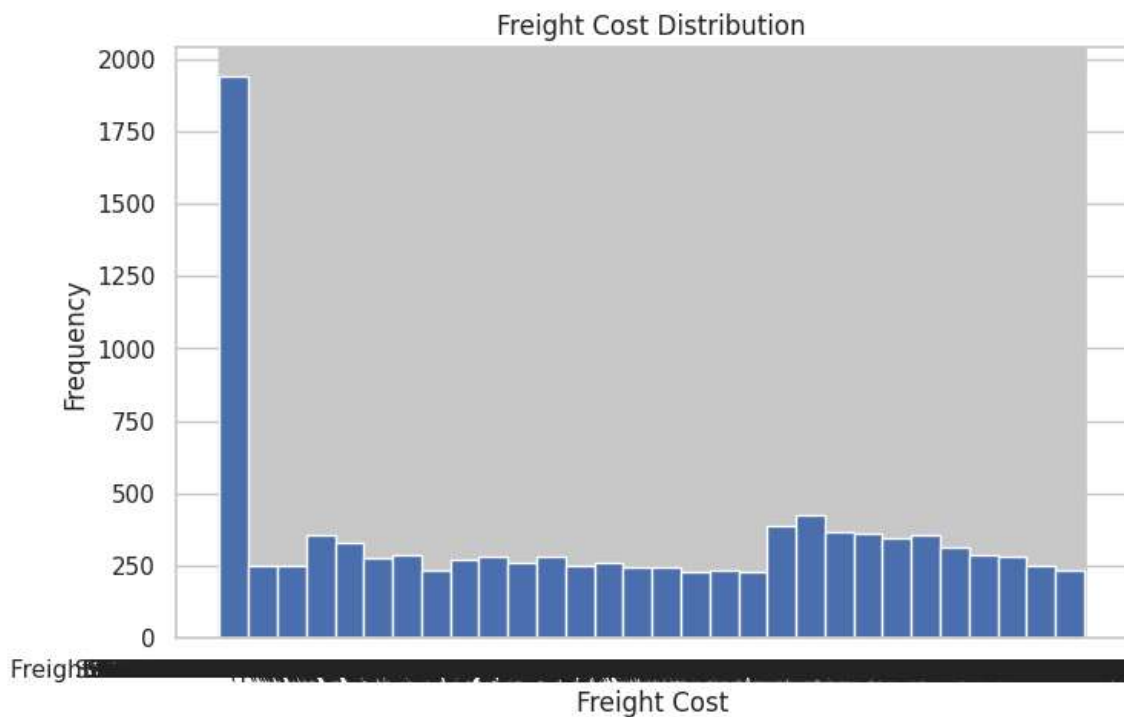
- 3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

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- ✓ Chart - 5

```
plt.figure(figsize=(8,5))
plt.hist(df['Freight Cost (USD)'], bins=30)
plt.title('Freight Cost Distribution')
plt.xlabel('Freight Cost')
plt.ylabel('Frequency')
plt.show()
```



- ✓ 1. Why did you pick the specific chart?

To understand how freight costs are distributed.

✓ 2. What is/are the insight(s) found from the chart?

Shows weather most shipments have low, medium or high freight costs.

✓ 3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Useful for transportation cost optimization and budgeting.

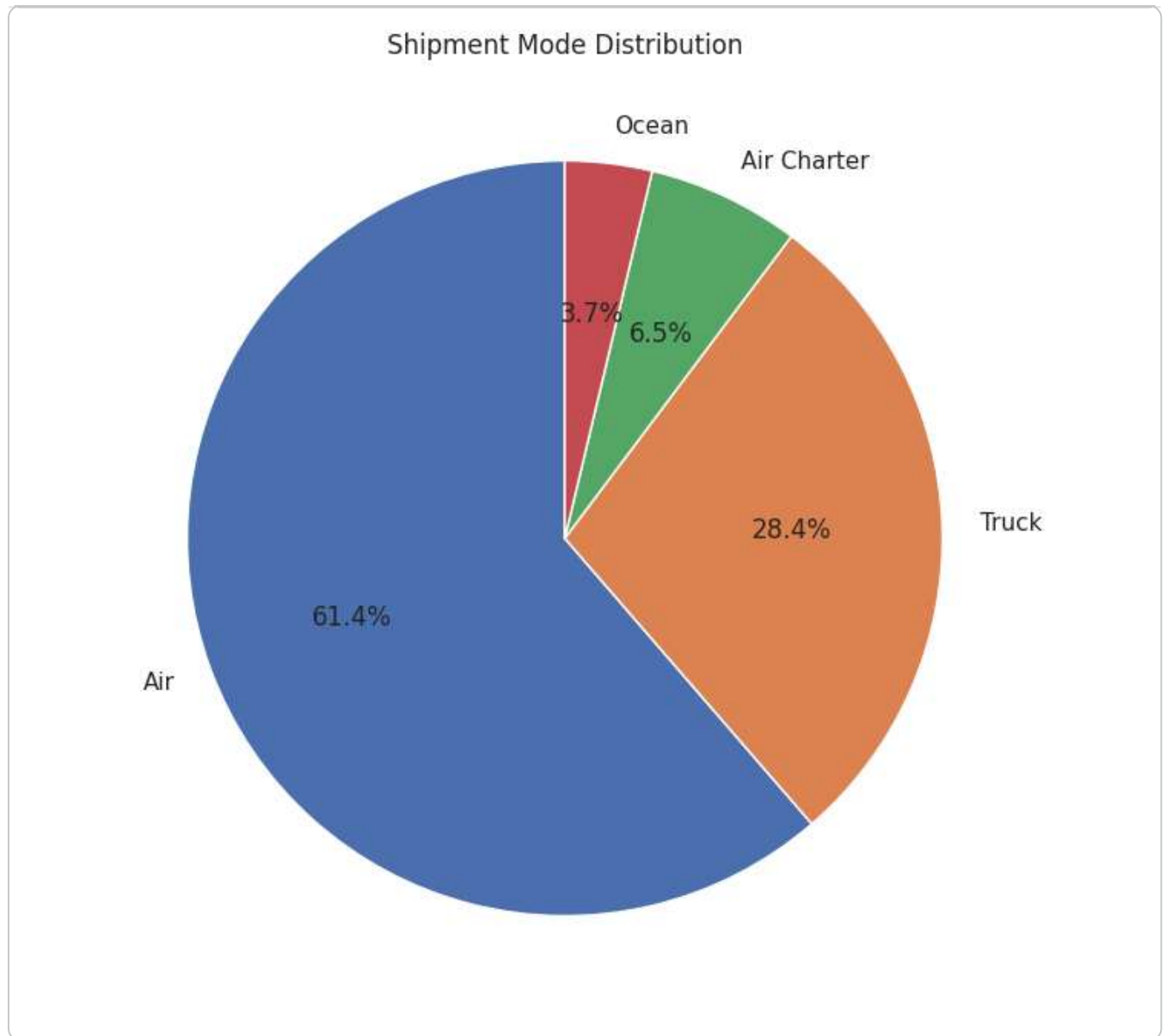
✓ Chart - 6

```
import matplotlib.pyplot as plt

shipment_mode = df['Shipment Mode'].value_counts()

plt.figure(figsize=(8,8))
plt.pie(shipment_mode,
        labels=shipment_mode.index,
        autopct='%1.1f%%',
        startangle=90)

plt.title('Shipment Mode Distribution')
plt.show()
```



- ✓ 1. Why did you pick the specific chart?

A pie chart effectively illustrates the proportion of each shipment mode.

- ✓ 2. What is/are the insight(s) found from the chart?

- One shipment mode account for the largest share of deliveries.
- Alternative shipment methods are used less frequently.

- 3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

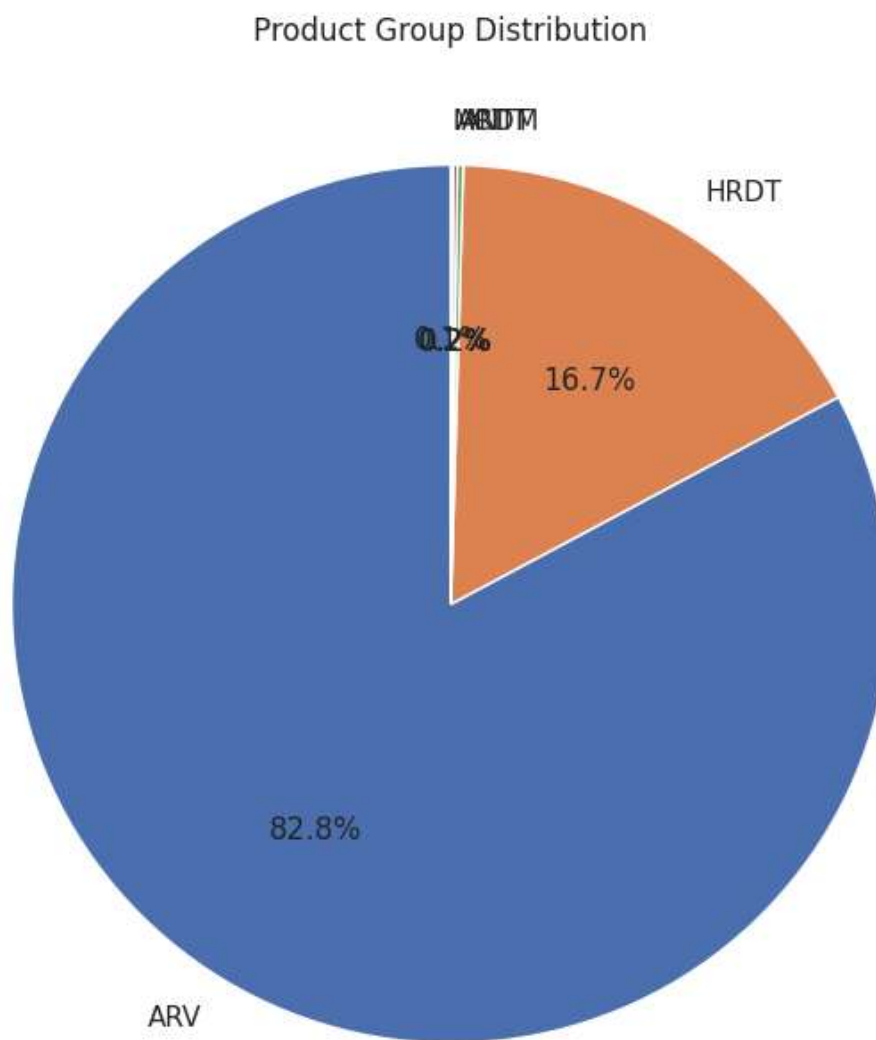
↳ 1 cell hidden

✓ Chart - 7

```
product_group = df['Product Group'].value_counts()

plt.figure(figsize=(8,8))
plt.pie(product_group,
        labels=product_group.index,
        autopct='%1.1f%%',
        startangle=90)

plt.title('Product Group Distribution')
plt.show()
```



✓ 1. Why did you pick the specific chart?

To visualize each product group's contribution to total shipments.

- ✓ 2. What is/are the insight(s) found from the chart?
 - A few product groups account for most deliveries.
 - Product demand is concentrated rather than evenly distributed.
- ✓ 3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Supports strategic product planning and efficient stock allocation.

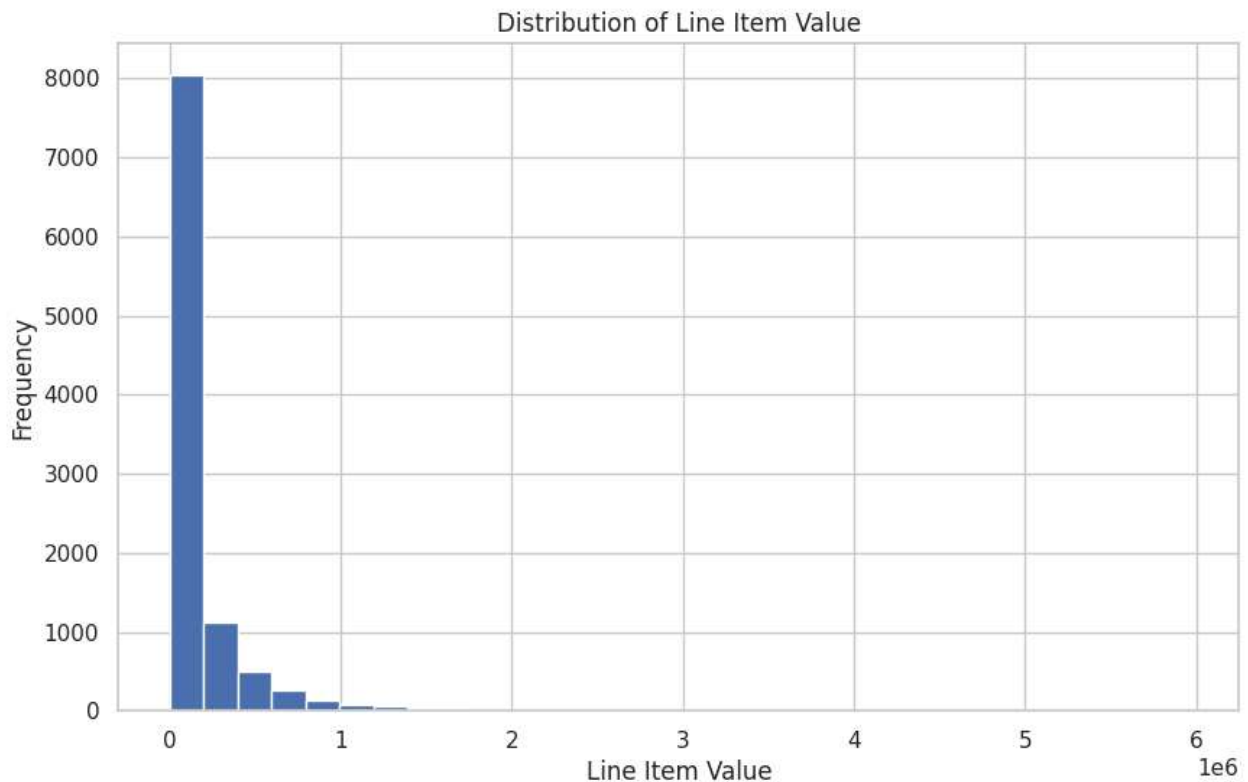
✓ Chart - 8

```
plt.figure(figsize=(10,6))

plt.hist(df['Line Item Value'],
        bins=30)

plt.title('Distribution of Line Item Value')
plt.xlabel('Line Item Value')
plt.ylabel('Frequency')

plt.show()
```



- ✓ 1. Why did you pick the specific chart?

To understand how shipment values are distributed.

✓ 2. What is/are the insight(s) found from the chart?

- Most shipments fall within a specific value range.
- Very high-value shipments are comparatively rare.

✓ 3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Helps identify pricing trends, revenue concentration, and high-value transactions requiring additional monitoring.

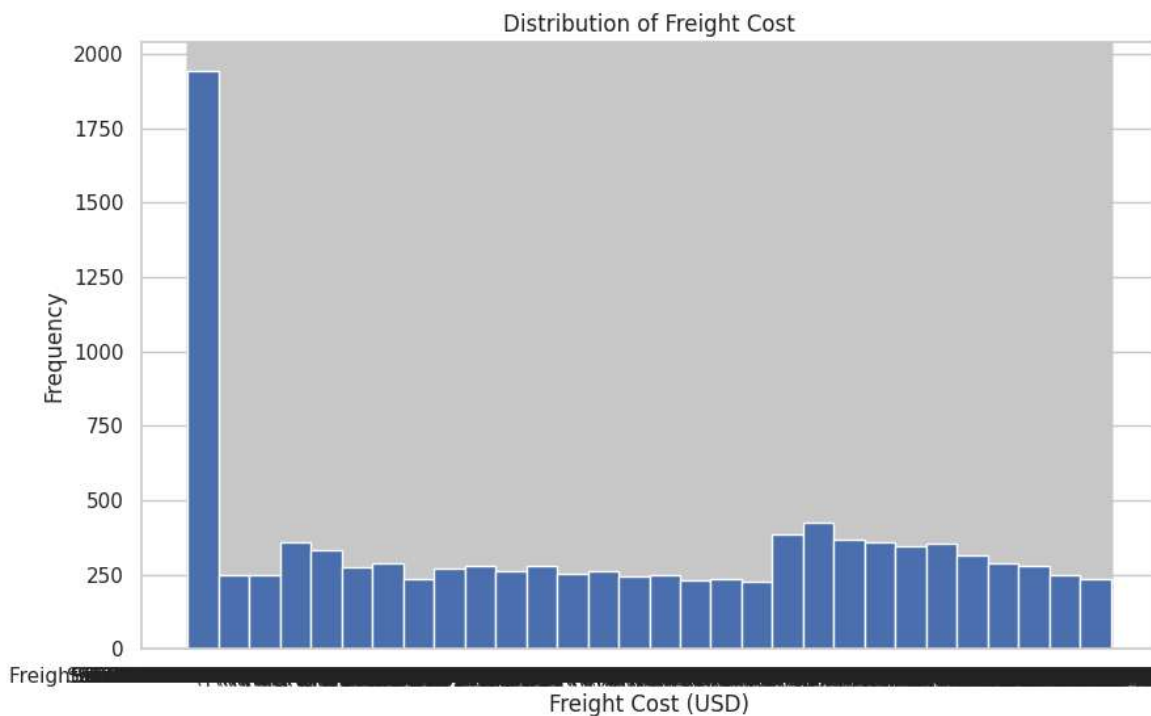
✓ Chart - 9

```
plt.figure(figsize=(10,6))

plt.hist(df['Freight Cost (USD)'],
        bins=30)

plt.title('Distribution of Freight Cost')
plt.xlabel('Freight Cost (USD)')
plt.ylabel('Frequency')

plt.show()
```



✓ 1. Why did you pick the specific chart?

To analyze how transportation costs are distributed.

✓ 2. What is/are the insight(s) found from the chart?

- Most shipments incur moderate freight costs.
- Only a small number have exceptionally high freight expenses.

✓ 3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Enable cost optimization and identification of unusually expensive shipments.

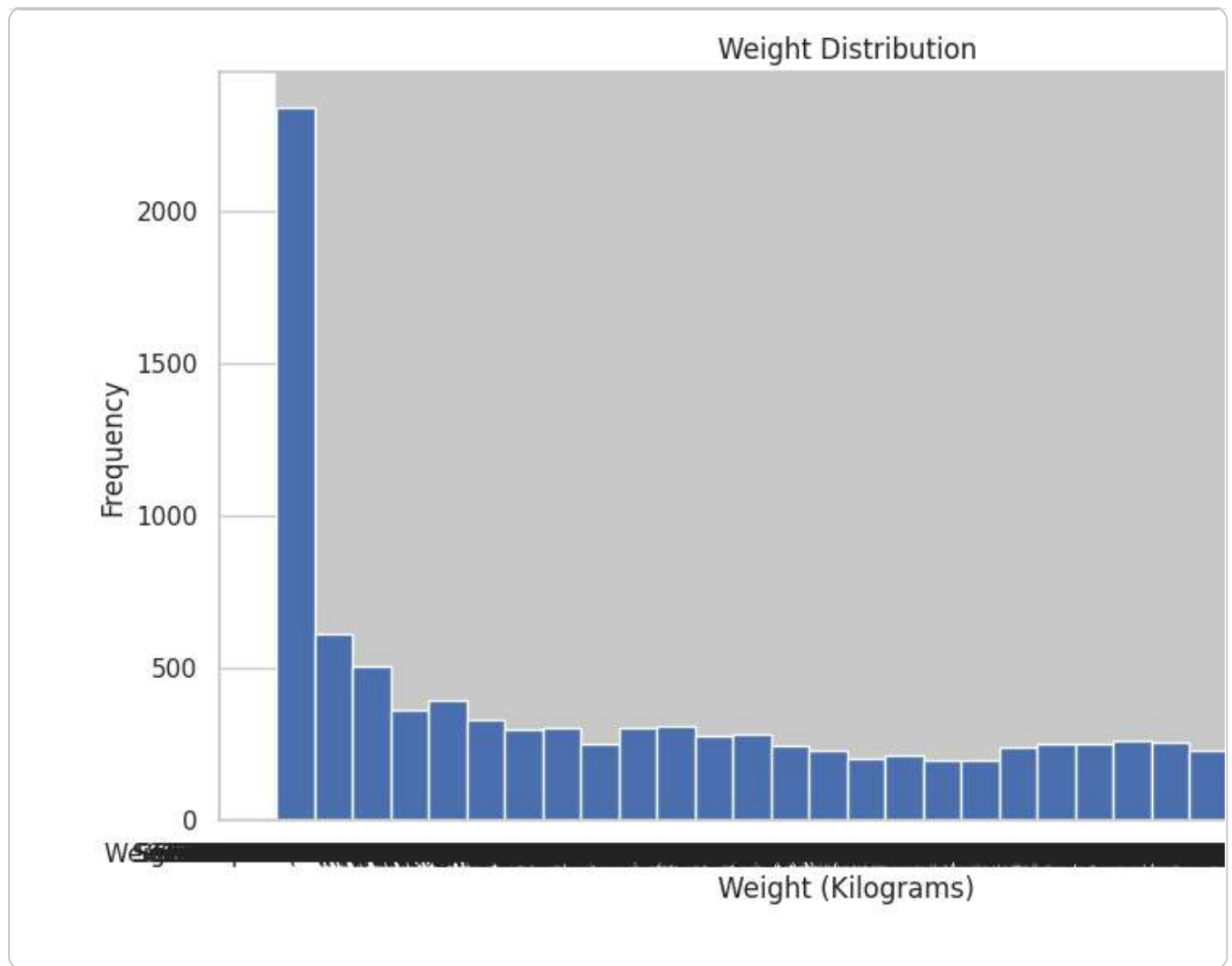
✓ Chart - 10

```
plt.figure(figsize=(10,6))

plt.hist(df['Weight (Kilograms)'],
        bins=30)

plt.title('Weight Distribution')
plt.xlabel('Weight (Kilograms)')
plt.ylabel('Frequency')

plt.show()
```



- ✓ 1. Why did you pick the specific chart?

To understand nshipment weight patterns.

- ✓ 2. What is/are the insight(s) found from the chart?

- Most shipments are concentrated within a moderate weight range.
- Heavy shipments occurs less frequently.

- ✓ 3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Improves transportation planning,vehicle utilization,and freight cost estimation.

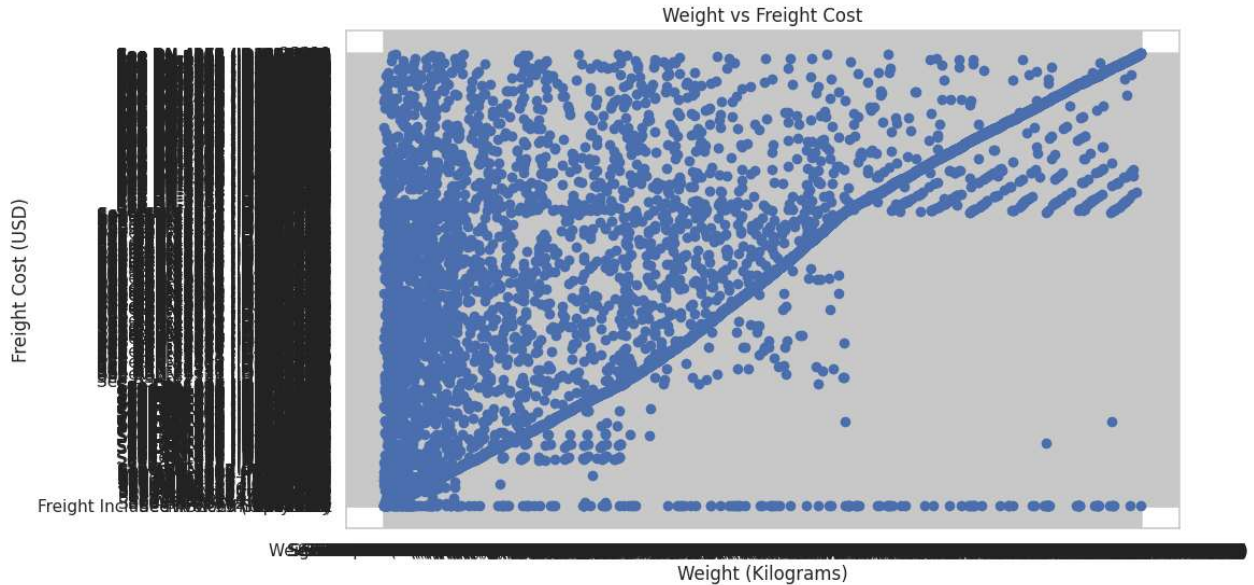
- ✓ Chart - 11

```
plt.figure(figsize=(10,6))
```

```
plt.scatter(df['Weight (Kilograms)'],
            df['Freight Cost (USD)'])

plt.title('Weight vs Freight Cost')
plt.xlabel('Weight (Kilograms)')
plt.ylabel('Freight Cost (USD)')

plt.show()
```



✓ 1. Why did you pick the specific chart?

A scatter plot is appropriate for examining the relationship between shipment weight and freight cost.

✓ 2. What is/are the insight(s) found from the chart?

- Freight cost generally increase as shipment weight increases.
- Some heavy shipments have unusually high transportation costs.

✓ 3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Support optimization of shipping expenses and identification of costly logistics operations.

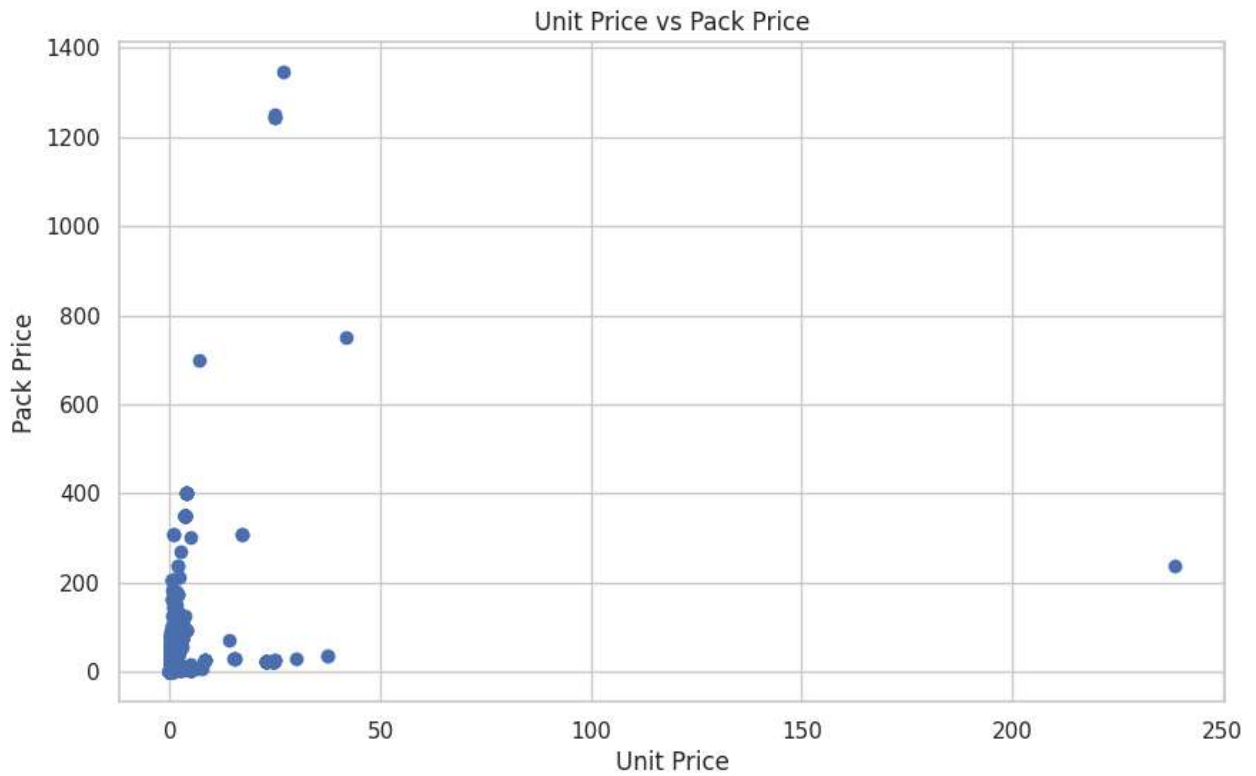
✓ Chart - 12

```
plt.figure(figsize=(10,6))

plt.scatter(df['Unit Price'],
            df['Pack Price'])
```

```
plt.title('Unit Price vs Pack Price')
plt.xlabel('Unit Price')
plt.ylabel('Pack Price')

plt.show()
```



- ✓ 1. Why did you pick the specific chart?

To examine the relationship between unit price and pack price.

- ✓ 2. What is/are the insight(s) found from the chart?

- Higher unit prices correspond to higher pack prices.
- A few products deviate from the general pricing trend.

- ✓ 3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Helps evaluate pricing consistency and identify products requiring pricing review.

- ✓ Chart - 13

```

top_vendor = df['Vendor'].value_counts().head(10)

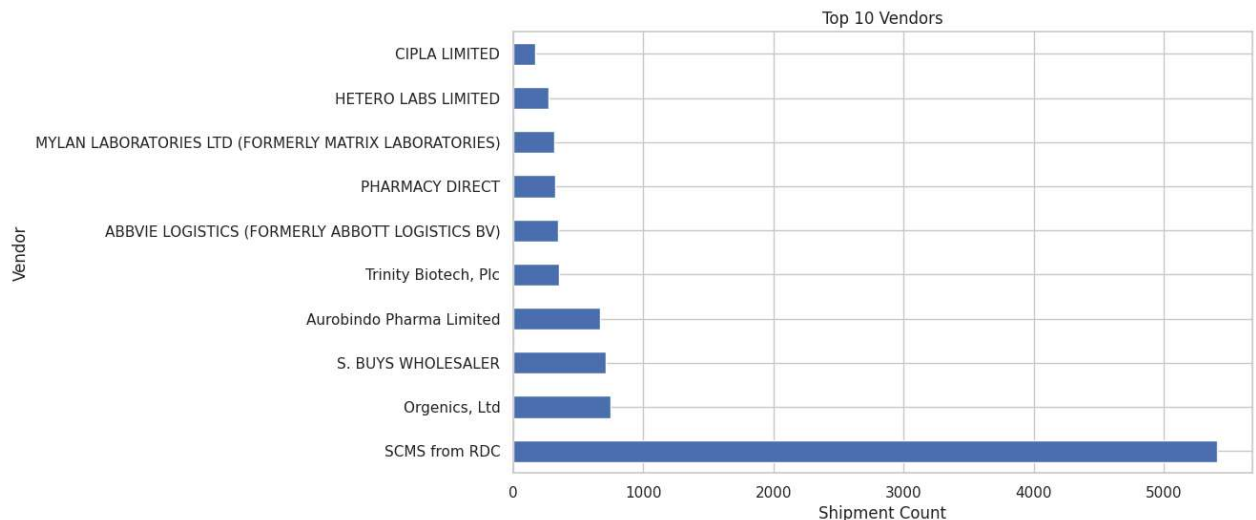
plt.figure(figsize=(10,6))

top_vendor.plot(kind='barh')

plt.title('Top 10 Vendors')
plt.xlabel('Shipment Count')

plt.show()

```



✓ 1. Why did you pick the specific chart?

A horizontal bar chart improves readability when comparing vendor names.

✓ 2. What is/are the insight(s) found from the chart?

- Vendor shipment volumes vary significantly.
- A small number of vendors contribute a substantial portion of shipment.

✓ 3. Will the gained insights help creating a positive business impact?

Are there any insights that lead to negative growth? Justify with specific reason.

Supports vendor performance evaluation and supplier management decision.

✓ Chart - 14 - Correlation Heatmap

```

top_site = df['Manufacturing Site'].value_counts().head(10)

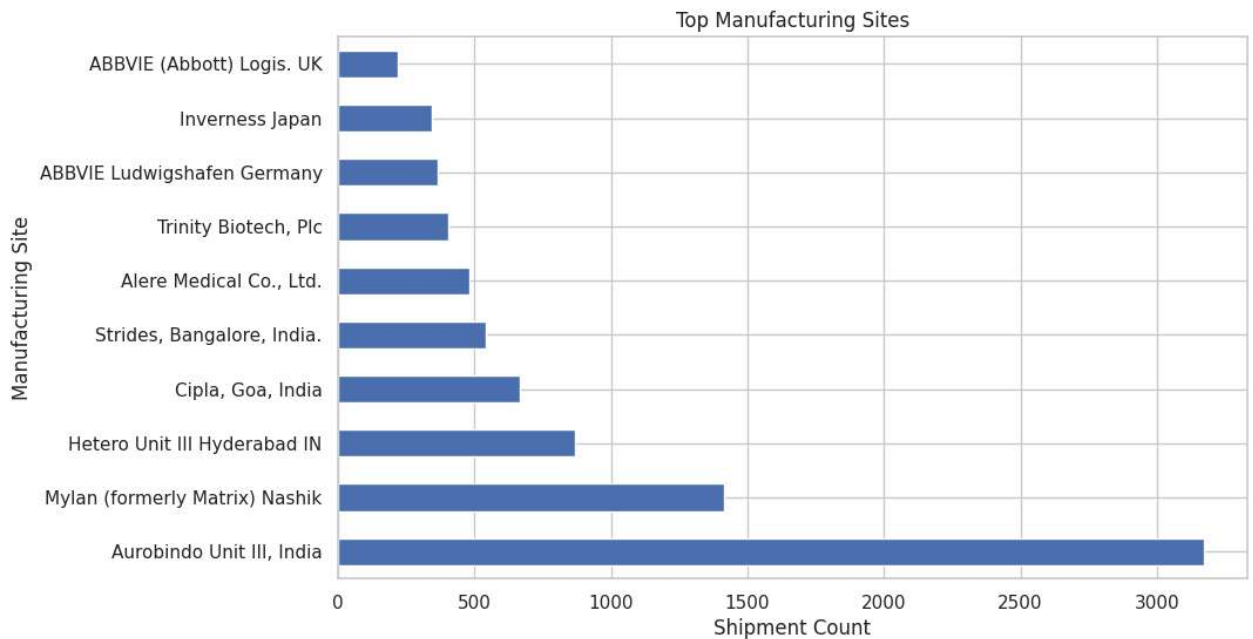
plt.figure(figsize=(10,6))

top_site.plot(kind='barh')

```

```
plt.title('Top Manufacturing Sites')
plt.xlabel('Shipment Count')

plt.show()
```



✓ 1. Why did you pick the specific chart?

To compare shipment output among manufacturing facilities with better label readability.

✓ 2. What is/are the insight(s) found from the chart?

- Certain manufacturing sites contribute significantly more shipments.
- Production capacity differs across facility.

✓ Chart - 15 - Pair Plot

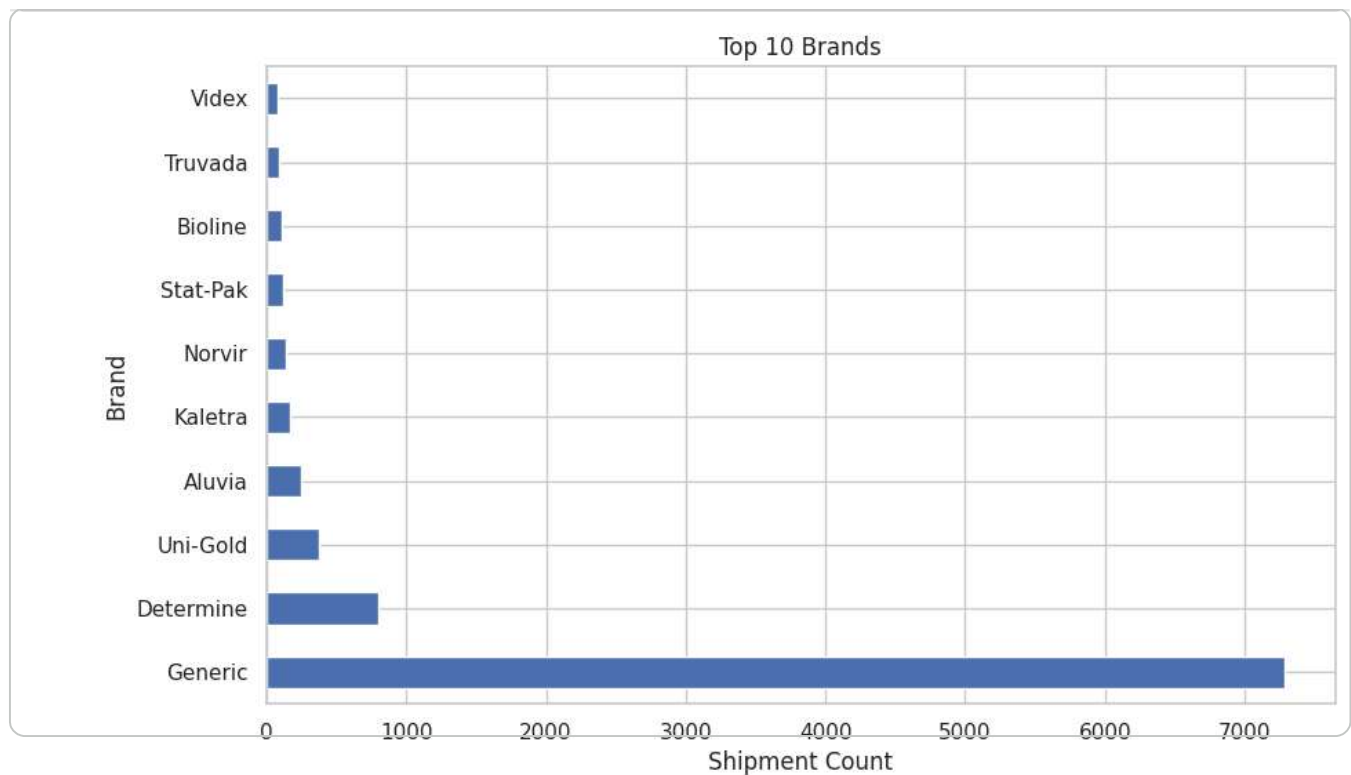
```
top_brand = df['Brand'].value_counts().head(10)

plt.figure(figsize=(10,6))

top_brand.plot(kind='barh')

plt.title('Top 10 Brands')
plt.xlabel('Shipment Count')

plt.show()
```



- ✓ 1. Why did you pick the specific chart?

To compare shipment volumes across brands using an easily interpretable format.

- ✓ 2. What is/are the insight(s) found from the chart?

- Some brands consistently outperform others.
- Brand demands varies considerably.